Total No. of Questions : 8]

SEAT No. :

PC2376

[Total No. of Pages : 2

[6354]-493
B.E. (Computer Engineering)
INFORMATION RETRIEVAL
(2019 Pattern) (Semester - VII) (Elective - IV) (410245 A)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) List out the General-Purpose Data Compression techniques and explain with suitable example. **[6]**

b) Describe in detail Invalidation List, Garbage Collection, Document Modifications in document deletion. **[6]**

c) Write a short note on. **[6]**

i) Modeling and coding

ii) Arithmetic Coding

OR

Q2) a) Explain software Architecture of the IR system in detail. **[6]**

b) Explain in details: **[6]**

i) Decoding performance

ii) Document Reordering

c) Describe data compression with Huffman coding. **[6]**

Q3) a) Explain categorization and filtering with any two detailed examples. **[7]**

b) Describe passage retrieval and ranking with example. **[5]**

c) Explain the Information-Theoretic Model in detail. **[5]**

OR

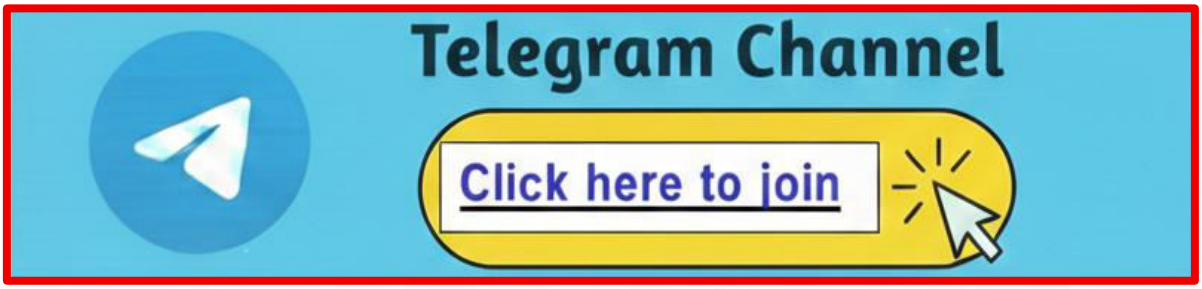
Q4) a) Explain probabilistic Classifiers & Generalized Linear Models. **[7]**

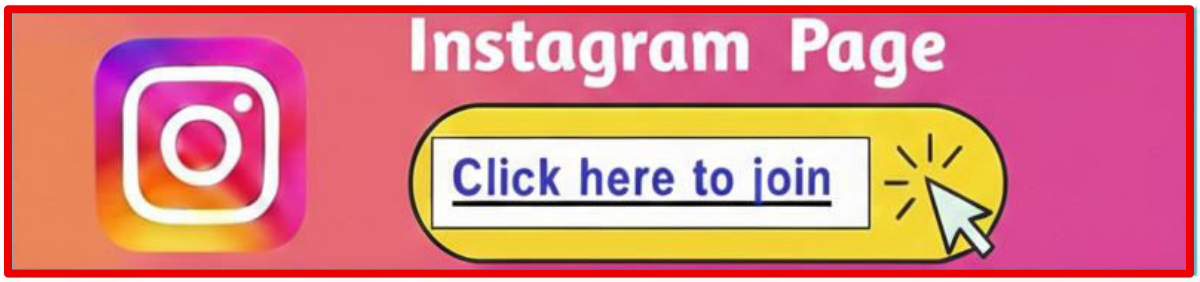
b) Explain Relevance Feedback Technique with suitable diagram. **[5]**

c) Describe Language Models and Smoothing. **[5]**

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Q5) a) Explain Measuring effectiveness like Traditional effectiveness measure and the text retrieval conference (TREC) with suitable examples. **[6]**

b) Write a short note on: **[6]**

i) Nontraditional effectiveness measures.

ii) Measuring efficiency

c) Explain query scheduling with suitable examples. **[6]**

OR

Q6) a) Write a short note on: **[6]**

i) Minimizing adjudication effort.

ii) Using statistics in evaluation

b) Explain caching with suitable examples. **[6]**

c) Differentiate between Redis and Memcached. **[6]**

Q7) a) Describe Map reduce with suitable examples. **[6]**

b) Write a short note on: **[6]**

i) The structure of the web

ii) Python Scrapy

c) Describe web crawler with its components. **[5]**

OR

Q8) a) Describe Parallel Query Processing with suitable examples. **[6]**

b) Explain the following term:

i) Static ranking

ii) Dynamic ranking **[6]**

c) Write a short note on Evaluation web search. **[5]**

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Total No. of Questions : 8]

SEAT No. :

P-6559

[Total No. of Pages : 2

[6181]-109

Fourth Year B.E. (Computer Engineering)

INFORMATION RETRIEVAL

(2019 Pattern) (Semester - VII) (410245 A) (Elective - IV)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) ***Solve Q.1 or Q.2 and Q.3 or Q.4 and Q.5 or Q.6 and Q.7 or Q.8.***
2) ***Figures to the right indicate full marks.***
3) ***Neat diagram must be drawn whenever necessary.***
4) ***Assume suitable data if necessary.***

Q1) a) What is Data Compression? Explain Modeling and Coding. **[9]**

b) Write a short note on : **[9]**

i) Invalidation List

ii) Garbage Collection

iii) Document Modifications

OR

Q2) a) Write a short note on : **[9]**

i) Decoding Performance

ii) Document Reordering

iii) Arithmetic Coding

b) Differentiate between Contiguous Inverted Lists & Noncontiguous Inverted. **[5]**

c) Differentiate between Non parametric Gap Compression & Parametric Gap Compression. **[4]**

Q3) a) Explain Categorization and Filtering with any two detailed Examples of each. **[8]**

b) Write a short note on Generalized Linear Models. **[9]**

OR

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Q4) a) Explain Probabilistic Classifiers, Information-Theoretic Model in detail. **[9]**

b) Describe Language Models and Smoothing. **[8]**

Q5) a) Explain Traditional effectiveness measure and The Text Retrieval Conference (TREC) with suitable examples. **[9]**

b) Write a short note on : **[9]**

i) Nontraditional effectiveness measures

ii) Measuring efficiency

OR

Q6) a) What is Scheduling and Caching in Measuring Efficiency?. Explain in detail. **[9]**

b) Write a short note on : **[9]**

i) Using statistics in evaluation

ii) Minimizing adjudication Effort

Q7) a) Explain in detail Parallel Query Processing with suitable examples. **[9]**

b) Write a short note on : **[8]**

i) Static ranking

ii) Dynamic ranking

OR

Q8) a) Describe MapReduce with suitable examples. **[9]**

b) Write a short note on : **[8]**

i) Evaluation web search

ii) Web Crawlers

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Total No. of Questions : 8]

SEAT No. :

P553

[Total No. of Pages : 2

[6004]-488
B.E. (Computer Engineering)
INFORMATION RETRIEVAL
(2019 Pattern) (Semester - VII) (Elective - IV) (410245 (A))

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Figures to the right side indicate full marks.***
3) ***Neat diagram must be drawn wherever necessary.***
4) ***Assume suitable data if necessary.***

Q1) a) Enlist the General-Purpose Data Compression techniques and explain with suitable examples. **[9]**

b) Write a short note on. **[9]**

i) Nonparametric Gap Compression.

ii) Parametric Gap Compression.

iii) Context-Aware Compression Methods.

OR

Q2) a) Write a short note on. **[9]**

i) Modeling and Coding.

ii) Huffman Coding.

iii) Arithmetic Coding.

b) Describe Dynamic Inverted Indices like Incremental Index Updates, Contiguous Inverted Lists and Noncontiguous Inverted. **[9]**

Q3) a) Explain Categorization and Filtering with any two detailed Examples. **[9]**

b) Explain the Information-Theoretic Model in detail. **[8]**

OR

Q4) a) Explain probabilistic Classifiers & Generalized Linear Models. **[9]**

b) Describe Language Models and Smoothing. **[8]**

Q5) a) Explain MeasuringEffectiveness like Traditional effectiveness measure and the text retrieval conference (TREC) with suitable examples. **[9]**

b) Write a short note on: **[9]**

i) Nontraditional effectiveness measures.

ii) Measuring efficiency.

OR

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Q6) a) Explain Query Scheduling and Caching with suitable examples. **[9]**

b) Write a short note on: **[9]**

i) Using statistics in evaluation.

ii) Minimizing adjudication Effort.

Q7) a) Describe Parallel Query Processing with suitable examples. **[8]**

b) Write a short note on: **[9]**

i) The structure of the web.

ii) Quires and Users.

iii) Static ranking.

OR

Q8) a) Describe Map Reduce with suitable examples. **[9]**

b) Write a short note on: **[8]**

i) Evaluation web search.

ii) Web Crawlers.

iii) Web crawler libraries.

iv) Dynamic ranking.

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Total No. of Questions : 8]

SEAT No. :

PA-918

[Total No. of Pages : 2

[5927]-350

B.E. (Computer Engineering)

INFORMATION RETRIEVAL

(2019 Pattern) (Semester - VII) (410245 A) (Elective - IV)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) ***Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Figures to the right indicate full marks.***
3) ***Neat diagrams must be drawn wherever necessary.***
4) ***Assume sitable data, if necessary.***

Q1) a) Explain Data Compression with Huffman Coding. **[9]**

b) What are the Context-Aware Compression Methods? Explain it in detail. **[9]**

OR

Q2) a) Write a short note on : **[9]**

i) Decoding Performance

ii) Document Reordering

iii) Arithmetic Coding

b) Explain in detail Invalidation List, Garbage Collection, Document Modifications in document deletion. **[9]**

Q3) a) Explain Categorization and Filtering with any two detailed Examples. **[9]**

b) Explain the Information-Theoretic Model in detail. **[8]**

OR

Q4) a) Explain Probabilistic Classifiers & Generalized Linear Models. **[9]**

b) Describe Language Models and Smoothing. **[8]**

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Q5) a) Explain Traditional effectiveness measure and The Text Retrieval Conference (TREC) with suitable examples. **[9]**

b) Write a short note on : **[9]**

i) Nontraditional effectiveness measures

ii) Measuring efficiency

OR

Q6) a) What is Scheduling and Caching in Measuring Efficiency?. Explain in detail. **[9]**

b) Write a short note on : **[9]**

i) Using statistics in evaluation

ii) Minimizing adjudication Effort

Q7) a) Describe Parallel Query Processing with suitable examples. **[8]**

b) Write a short note on : **[9]**

i) The structure of the web

ii) Quires and Users

iii) Static ranking

OR

Q8) a) Describe MapReduce with suitable examples. **[9]**

b) Write a short note on : **[8]**

i) Evaluation web search

ii) Web Crawlers

iii) Web crawler libraries

iv) Dynamic ranking

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Total No. of Questions : 8]

SEAT No. :

PD4584

[Total No. of Pages : 2

[6404]-89

B.E. (Computer Engineering)

INFORMATION RETRIEVAL

(2019 Pattern) (Semester - VII) (410245A) (Elective - IV)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Question No. 1 or 2 and 3 or 4 and 5 or 6 and 7 or 8.***
2) ***Neat diagram must be drawn whenever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) Describe a Huffman coding. **[6]**

b) If you want to reduce total size of index what you would focus on justify your answer. **[6]**

c) How to measure performance and what are the ways for Index Compression for High Query Performance. **[6]**

OR

Q2) a) Explain the Arithmetic coding with suitable example. **[6]**

b) Explain Dynamic Inverted Indices with suitable example. **[6]**

c) Differentiate between Contiguous Inverted Lists and Non-contiguous Inverted. **[6]**

Q3) a) Illustrate Generalized Linear Models. **[7]**

b) What is Language Modelling explain Related Methods. **[5]**

c) When estimating the probabilities in language modelling what are the problems observed and how to resolve it? **[5]**

OR

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Q4) a) Explain The Binary Independence Model in detail. **[7]**

b) Why BM25 is better that TF-TDF? **[5]**

c) Explain the smoothing probability estimates. **[5]**

Q5) a) What is precision and recall? **[6]**

b) What are the Characteristic of TREC collection. **[6]**

c) Write the types of performance measure optimization. **[6]**

OR

Q6) a) What are the challenges with online evaluation? **[6]**

b) What are the Efficiency measures in information retrieval? **[6]**

c) Differentiate between Redis and Memcached. **[6]**

Q7) a) Examine various ways of making information retrieval systems scale to very large text collections such as the Web. **[6]**

b) Write Basic PageRank algorithm. **[6]**

c) Draw and explain Components of a Web crawler. **[5]**

OR

Q8) a) Explain Map reduce basic framework. **[6]**

b) How to evaluate a web search. **[6]**

c) Differentiate between Python Scrapy and Beautiful Soup. **[5]**



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Total No. of Questions : 8]

SEAT No. :

PE-2200

[Total No. of Pages : 2

[6584]-99
B.E. (Computer Engineering)
INFORMATION RETRIEVAL
(2019 Pattern) (Semester - VII) (410245 A) (Elective - IV)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) *Solve Q1 or Q2 and Q3 or Q4 and Q5 or Q6 and Q7 or Q8.*
2) *Figures to the right indicates full marks.*
3) *Neat diagram must be drawn whenever necessary.*
4) *Assume suitable data if necessary.*

Q1) a) Explain General purpose Data Compression for High Query Performance in detail. **[6]**

b) List out various Huffman Coding Techniques and explain Length Limited Huffman code technique in detail. **[6]**

c) Explain Parametric Gap Compression in detail. **[5]**

OR

Q2) a) Explain different possible classes of symbol wise compression in detail. **[6]**

b) Which are the different Context-Aware Compression Methods, explain any Context- Aware Compression Method in detail. **[6]**

c) Differentiate between Huffman coding versus arithmetic coding. **[5]**

Q3) a) Explain the concept of term frequency in detail. **[6]**

b) Give in detail the concept of Language Models and Smoothing. **[6]**

c) Enlist various simple document features and explain any one features in detail. **[6]**

OR

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Q4) a) Explain Document Length: BM25 retrieval system in detail. **[6]**

b) Illustrate the process of Relevance Feedback in detail with suitable example. **[6]**

c) Explain the various problems of categorization and filtering. **[6]**

Q5) a) List and explain different aspects of IR evaluation. **[6]**

b) Explain Throughput and Latency used while evaluating performance of a search engine strategies in detail. **[6]**

c) What is the role of Minimizing Adjudication Effort to produce a set of judgments for IR evaluation. **[5]**

OR

Q6) a) Differentiate between Arithmetic Mean Versus Geometric Mean in detail. **[6]**

b) Which are the different methods are used for Selecting the Documents for Adjudication. **[6]**

c) Elaborate the use of Statistical analysis to estimate a system's performance. **[5]**

Q7) a) Which are the different ways used by search engine to process the queries faster, explain with suitable example. **[6]**

b) Explain the concept Term Partitioning in detail with suitable example. **[6]**

c) Explain the term Dynamic Ranking in detail. **[6]**

OR

Q8) a) Explain the concept of Extended PageRank in detail. **[6]**

b) Elaborate the Various categories of user needs in web queries for query analysis. **[6]**

c) Brief in detail the structure of Web. **[6]**

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